

Quantifying Divergence for Human-AI Collaboration and Cognitive Trust

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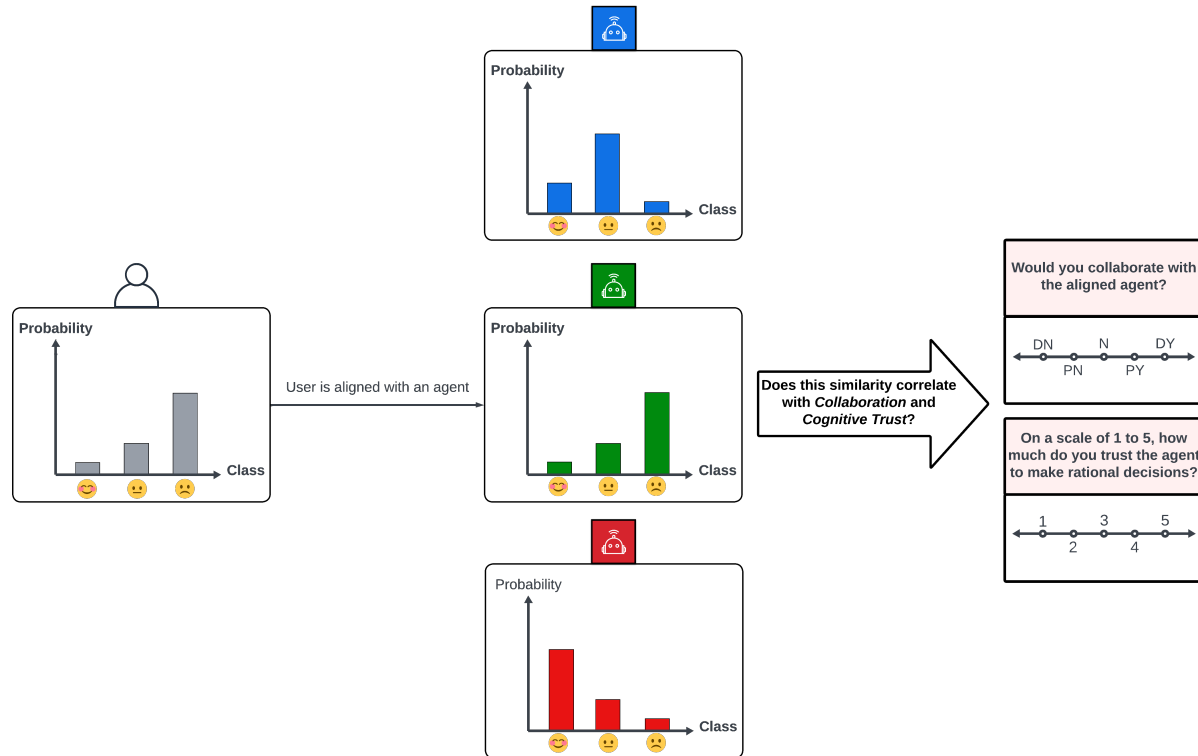


Figure 1: Class probabilities from three models (Blue, Green, Red) showing soft labels for three classes in the Natural Language Inference (NLI) task: Entailment (Smiley Face), Neutral (Neutral Face), and Contradiction (Sad Face). The user is aligned with the most similar model (Green) using divergence metrics. Two key questions are asked: (1) “Would you collaborate with this aligned agent?” and (2) “On a scale of 1 to 5, how much do you trust this agent to make rational decisions?”. Responses to the collaboration question are recorded using a Likert scale: DN (Definitely No), PN (Probably No), N (Neutral), PY (Probably Yes), and DY (Definitely Yes). Cognitive trust is rated on a scale from 1 (lowest) to 5 (highest). This process examines the relationship between decision-making similarity, likelihood of collaboration, and cognitive trust in the aligned AI agent.

Abstract

Predicting the collaboration likelihood and measuring cognitive trust to AI systems is more important than ever. To do that, previous

research mostly focused solely on model features (e.g., accuracy, confidence) and ignored the human factor. To address that, we propose several *decision-making similarity* measures based on divergence metrics (e.g., KL, JSD) calculated over the labels acquired from humans and a wide range of models. We conduct a user study (N=100) on a textual entailment task, where the users are provided with soft labels from various models and asked to pick the closest option to them. The users are then shown the similarities/differences to their most similar model and are surveyed for their likelihood of collaboration and cognitive trust to the selected system. Finally, we qualitatively and quantitatively analyze the relation between

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the proposed *decision-making similarity* measures and the survey results. We find that people tend to collaborate with their most similar models—measured via JSD—yet this collaboration does not necessarily imply a similar level of cognitive trust. We release all resources related to the user study (e.g., design, outputs), models, and metrics at our repo ¹.

CCS Concepts

• **Computing methodologies** → **Machine learning**; **Natural language processing**; • **Human-centered computing** → **Human computer interaction (HCI)**.

Keywords

Human-AI Collaboration, Cognitive Trust, Decision-Making Similarity (DMS), Divergence Metrics, KL Divergence, JSD, Soft Labels

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1 Introduction

Human-AI interaction is becoming increasingly important in our daily lives as the number of AI assistants such as ChatGPT [30] grows. Humans employ these assistants in many different areas ranging from healthcare [19, 20, 26] to education [1, 15, 28]. In a survey conducted by Skjuve et al., authors find that individuals use ChatGPT with the motivations such as productivity, novelty, creative work, learning and development [36]. For instance, developers use these assistants to increase their productivity. Barke et al. highlight that developers use them ² both to expedite their coding processes and explore novel coding approaches [7]. People also use these assistants for producing creative or formal content, e.g., writing articles, stories or formal letters. Finally, they are more commonly utilized as knowledge repositories akin to search engines, e.g., how an air conditioner cools a room, to enhance the knowledge of their users. This increasing amount of interaction raises two critical questions: How much do we prefer to collaborate with AI models where we seek their assistance, and how much do we trust them? Predicting and measuring these before deploying AI models in real-life scenarios is crucial to foresee how the interaction will unfold. Therefore, understanding the collaboration and cognitive trust dynamics is crucial to build future AI assistants.

Previous works predominantly focus on predicting the likelihood of collaboration using model-only or user-only features. For instance, Vodrahalli et al. develop a human behavior model to predict when users will accept AI recommendations [39]. The study incorporates a range of features, including model confidence, user confidence, demographic information (e.g., age, sex, and education level), and user opinions for AI (e.g., how often AI aids in user's daily tasks). Yin et al. examine the impact of the stated and observed accuracy of the models during human-AI interactions on the trust of the users [44]. In addition to common model features

like accuracy, the effect of model generated explanations on human collaboration and trust [5, 21, 35, 45] are explored. On a similar line, Bansal et al., highlight the importance of model predictability, showing that when participants *know* when AI would succeed/fail, the level of collaboration increases [4]. In short, existing work either employ shallow (e.g., accuracy), or sophisticated (e.g., explanations, predictability) model features; along with shallow user features (e.g., age). However, studying user and model features *jointly* to *quantify* and *predict* the level of collaboration and cognitive trust to AI, is left unexplored to the best of our knowledge.

To address that, we propose a novel experimental setup sketched in Fig. 1 to analyze the *decision-making similarity* (DMS) between human and AI models and its impact on collaboration and cognitive trust. We first implement a diverse set of models ranging from statistical models to GPT-family ones specifically: tf-idf, Enhanced LSTM [12], RoBERTa [46], and davinci-003 [10] on Natural Language Inference (NLI) task. Then we conduct a multi-staged user study: In the first stage, we present the generated label distributions, a.k.a. *soft-labels*, from the models to participants and ask them to choose the closest label distribution; where in the second stage, users answer various questions regarding collaboration and cognitive trust to the *closest* model. We define the distance as the divergence between the human-AI predictions, i.e., soft labels, using various metrics (e.g., Jensen-Shannon Distance (JSD) [25], Kullback-Leibler (KL) [22]). Finally we analyze the relation between these metrics, a.k.a., DMS, and the user ratings for collaboration and cognitive trust.

Our initial findings suggest that: (1) People tend to collaborate with AI models when their decision-making processes—measured via DMS—are similar. However, they are less likely to develop cognitive trust in these models. Surprisingly, they may exhibit stronger collaboration tendencies with AI models even when their trust in the model is low. (2) A particular type of DMS, low inverse KL divergence, has the most influence on collaboration likelihood: agreeing on the same answer (right or wrong) with high confidence. (3) Finally, cognitive trust seems also related to low inverse-KL; however, it might also require another type of DMS with low forward-KL, i.e., avoiding being overconfident in case of uncertainty. Therefore, its likelihood can be indicated by JSD. It should be noted that, our findings are indeed encouraging, however, are bound to the small set of experimented models, task, dataset and user pool. With this work, we hope to pave the way towards incorporating the soft-labels to user studies and use the suggested divergence metrics to approximate collaboration and trust to AI models *before model deployment*.

Our contributions can be listed below:

- To the best of our knowledge, this is the first study that investigates the impact of “decision-making similarity” (DMS)—measurable via divergence metrics on soft labels—on human-AI collaboration and cognitive trust AI.
- We propose a comprehensive, yet, flexible four-staged user study to measure DMS. Although originally designed for the NLI task, it is easily adaptable to any classification task.

¹<https://github.com/gglab-ku/cogeval>

²<https://github.com/features/copilot>

- To encourage further studies on collaboration and trust, we share all resources, including the user study design, participant outputs, NLI models and predictions and related implementation.

For definitions of technical terms such as soft labels, 3GS, KL divergence, and JSD, please refer to Appendix A.

2 Task Setup

To link the dimensions of human-AI interaction with the model features, we first identify a downstream task that requires reasoning (see §2.1). Then we train a diverse set of models and acquire soft labels (see §2.2). Finally, we propose and calculate several metrics to calculate the distance between the labels provided by humans and models (see §3), which we refer to as *decision-making similarity*.

2.1 Dataset

The SNLI (Stanford Natural Language Inference) dataset [9] is a well-known resource used to evaluate the natural language inference capabilities of models. The task is mostly associated with reasoning abilities and is part of the popular language understanding benchmark [40]. Here, each instance is annotated by five workers, and the resulting premise-hypothesis pairs were labeled as 3GS, 4GS, or 5GS, as shown in Table 1, where the number indicates the maximum number of agreements. To reveal differences between models and create variation in the user study, we use 4GS pairs instead of the more trivial 5GS. Finally, we reduce the sample size from 90 to 50, considering the human attention span.³

2.2 Models

To diversify the model features—model *accuracy*, *confidence*, and *soft-label distribution*—we consider both neural and non-neural models, different architectures (e.g., autoregressive decoder and bidirectional encoder), and different techniques such as fine-tuning and zero-shot prompting. We particularly include models with the highest accuracy [12, 46], highest confidence [10], diverse label distribution and include a random baseline for comparison. All the models are trained on SNLI training split by *excluding* the instances that are demonstrated to human annotators. Detailed model descriptions are available in the Appendix B.

3 Similarity Calculation

We define the *decision-making similarity* as the difference between two probability distributions⁴, P and Q , measured by KL and Jensen-Shannon divergence. Here, P refers to the soft-label distributions from human annotations, whereas Q refers to the label probabilities from models, a.k.a. agents.

Forward and Inverse KL Divergence. KL divergence, denoted as $D_{KL}(P \parallel Q)$, quantifies the information loss when using distribution Q instead of P :

³Participants must carefully examine each model’s confidence score and conduct analyses on similarities and differences. The annotation of 50 instances takes around 60 minutes.

⁴We have also investigated various distance metrics mostly based on hard labels, such as minimum square error, agreement percentage and Pearson’s correlation. Due to the unpromising results and the lack of space in the paper, those are omitted.

$$\alpha\text{-KL} = D_{KL}(P \parallel Q) = \sum_{x \in \mathcal{X}} P(x) \log \left(\frac{P(x)}{Q(x)} \right) \quad (1)$$

We refer to the above divergence as the *forward-KL* and denote it with α -KL. One would expect a high α -KL when human annotations are more distributed, i.e., P is wider, but the model is overconfident on one label, i.e., Q is narrow. Similarly, the *inverse KL* divergence is given by:

$$\beta\text{-KL} = D_{KL}(Q \parallel P) = \sum_{x \in \mathcal{X}} Q(x) \log \left(\frac{Q(x)}{P(x)} \right) \quad (2)$$

β -KL is larger in the opposite scenario—when P is narrow, i.e., humans mostly agree on a label, but Q is wide, i.e., the model’s label probabilities are widely distributed, or Q has a peak on another label that humans do not agree on. Therefore, the α -KL is mainly influenced by instances with low human agreement but high model confidence. In contrast, the β -KL indicates situations where humans mostly agree, but the model lacks confidence. Both types of situations can reveal discrepancies in decision-making similarities, hence valuable for the user study.

Jensen-Shannon Divergence: Although similar to KL-divergence, it is symmetric and finite, which makes it more intuitive [29].

$$D_{JSD}(P \parallel Q) = \frac{1}{2} \left(D_{KL} \left(P \parallel \frac{P+Q}{2} \right) + D_{KL} \left(Q \parallel \frac{P+Q}{2} \right) \right) \quad (3)$$

We use Jensen-Shannon Distance calculated as $\sqrt{D_{JSD}}$ in our analysis. JSD considers the mixture distribution of P and Q and calculates forward KL divergences from both distributions. Consequently, one would expect JSD to reflect both α and β KL. Hence, we anticipate JSD to strike a balance, effectively capturing the impact of instances with high human agreement but low machine confidence and vice versa. Therefore we choose JSD as the primary distance metric for the user study.

4 User Study

We conduct a user study with 100 college-educated annotators with good command of English, recruited from Prolific⁵. We require annotators to have a minimum approval rate of 90%. Further, we have set the college level as a filtering criterion. We have eliminated the participants from 159 to 100. The median time taken for the study was 35 minutes. Our average payment for each annotation was £4.00⁶. Furthermore, we filtered out user results with an accuracy below 0.60 during the annotation phase.

Before presenting instances to the participants, we filter out cases where two or more models provided identical hard labels with a confidence score exceeding 0.95. Such instances are considered trivial because when multiple models are highly confident (above 95%) and in complete agreement on the label, the underlying soft label distributions show little variation. In these cases, the decision-making process is obvious, and there is minimal nuance for comparing the subtle differences between human judgments and model predictions. Removing these instances allows us to focus on more ambiguous or challenging examples, where variation in

⁵<https://app.prolific.co/>

⁶The minimum wage in the UK is £10.42/hr.

Premise	Hypothesis	Soft Labels	Hard Label	Type
Two kids playing on a street.	Two children have fun on a street.	E, E, E, N, N	E	3GS
A group of bikers pedal along.	A group of cyclists are in a marathon.	N, N, N, N, E	N	4GS
A man fishing in the ocean.	A man is driving a car.	C, C, C, C, C	C	5GS

Table 1: A sample of validated SNLI Pairs from [9]. E: Entailment, N: Neutral, C: Contradiction

the soft labels provides a richer signal for analyzing the relationship between divergence metrics (e.g., KL divergence, JSD) and user judgments of collaboration and trust.

This results in the removal of 43 out of 90 instances. As a result, the accuracy of the LSTM and RoBERTa models falls short of aggregated human performance. To maintain the accuracy ranking, we introduce an additional 7 instances where the models succeed while the aggregated human answer fails. The accuracy variation between two sets can be found in Table 2.

Model	Acc _{original/subset}
Random Baseline	0.35 / 0.26
TF-IDF	0.51 / 0.28
LSTM (ESIM)	0.84 / 0.72
RoBERTa	0.84 / 0.72
Davinci	0.59 / 0.42
Human	0.81 / 0.72

Table 2: Model accuracy variations with subset selection. Human: Aggregated human annotations.

Our user study comprises four stages: i) training, ii) quiz, iii) annotation, and iv) dynamic survey. Detailed explanations for each stage are provided in Appendix B.

5 Results

Our main results are given in Table 3. Here, the collaboration and cognitive trust scores are averaged over the users that are *aligned* with the model (shown with the rows). Then each score is again averaged to get a final score for each aspect. As given in Table 3, the majority of the users (60%) are aligned with RoBERTa, followed by LSTM (34%) and Davinci (6%). None of the users are found similar to the random baseline or the tf-idf model, validating the suitability of the chosen metric and the quality of the user study. We find that the average collaboration score (3.88/5) is considerably higher than the average cognitive trust (3.34/5) score. Finally, in Table 4, we provide additional information on the models, users and their decision variations. We elaborate on the results below separately for collaboration and cognitive trust.

Collaboration: According to Table 3, Davinci has the highest collaboration score despite having the lowest cognitive trust score. It has also the lowest β -KL divergence to its aligned users. As discussed in §3, β -KL gets higher when the user predicts a label with high confidence, but the model has less confidence. Therefore, Davinci most effectively fulfills confidence expectations of its aligned users for their answers. In contrast, users aligned with

the RoBERTa and LSTM models exhibit higher β -KL values. That means users are more confident than their models in their predictions. They also have lower collaboration ratings, suggesting that agreeing on the same answer with high confidence is a critical DMS feature for collaboration. As a result, as seen in Table 4, Davinci and its corresponding users establish the most confident aligned user-machine pairs.

Cognitive Trust: We expect participants to have a sense of the rational characteristics of the models after the user study. This is because the disagreement and agreement cases are shown exclusively to them during the dynamic evaluation phase. Our results in Table 3 shows that LSTM has the highest cognitive trust scores, followed closely by RoBERTa. However, we notice a significant decrease in trust for Davinci. As given in Table 3, Davinci suffers from high α -KL. As discussed in §3, α -KL gets higher when a model is overconfident in its predicted label, but the user has low confidence in that label; instead, user confidence spreads over the labels. Davinci has also relatively lower accuracy than the average accuracy of its aligned users as shown in Table 4. This also indicates the overconfidence of the model in wrong labels for most of the questions. We think that observing this dissimilarity between themselves and models damages cognitive trust of users in the machine. We further observe that, JSD, where combines the effects of both α -KL and β -KL divergences give the same model rankings (LSTM-RoBERTa-Davinci) for cognitive trust. Therefore, it might be more advantageous to incorporate both divergences and solely rely on JSD for assessing cognitive trust. However more comprehensive model comparisons are needed to draw a concrete solution, which is not feasible in this user study.

To summarize, we observe that participants tend to collaborate with their most similar models; and they give higher collaboration ratings when they share high confidence in answers with their models, as indicated by **lower β -KL**. However, this is not the case for cognitive trust. Participants give higher trust ratings when they share low confidence in their answers and avoid being overconfident, as indicated by **lower α -KL** and **JSD**. The results also demonstrate that individuals may assign higher collaboration ratings to models, even if they have low cognitive trust to the model. Distributions of the collaboration and cognitive trust can be found in Appendix D.

6 Discussion and Analysis

(1) When multiple models share the same final labels, which one do people typically choose? We analyze the test instances where multiple models provide the same hard labels but distinct label distributions. We then compare how frequently the users choose the highly confident model versus the less confident model.

#p	Aligned model	α -KL	β -KL	JSD	Collaboration Rating	Cognitive Trust Rating
60	RoBERTa	0.53	1.21	0.23	3.9	3.36
34	LSTM	0.64	1.20	0.22	3.7	3.41
6	Davinci	0.81	0.98	0.26	4.1	2.66
100	average	0.64	1.19	0.23	3.88	3.34

Table 3: User study rating results over 5 points. #p: Number of participants aligned with the model. Rows show the averaged scores for the model calculated over the users aligned with it. The final row shows the averaged scores over all models.

Aligned Model	Model Accuracy	User Accuracy	Model Confidence	User Confidence	Soft Label Agreement	Hard Label Agreement	Selected Higher Confidence	Selected Lower Confidence
RoBERTa	0.72	0.69	0.80	0.86	0.25	0.71	0.40	0.05
LSTM	0.72	0.70	0.83	0.84	0.39	0.69	0.16	0.14
Davinci	0.42	0.66	0.94	0.91	0.53	0.64	0.05	0.05

Table 4: User-model performances and selection details. Soft label agreement: Number of times user-model agree on soft label. Hard label agreement: Number of times user-model agree on hard label. Selected higher/lower confidence: Number of times users selecting different model with higher/lower confidence on the same label.

The majority of participants (67%) choose the most confident model answer for 97% of the cases. This result supports the findings of Vodrahalli et al., which suggests individuals accept solutions from highly confident models [39]. We also see that as average user accuracy decreases, users tend to choose alternative models with the same label but lower confidence. So their answers are distributed more evenly across other labels. To quantify, we calculate Pearson’s r between the number of times users selected the models and the average user accuracy on that question. With $r = -0.41$, the number of times selecting low confident models increases as average user accuracy decreases. This result suggests that humans will often choose the most confident model, but the label distribution of a model will also be critical in cases when they are uncertain.

(2) When do people not choose their most similar model?

We first calculate the number of times the user directly chooses its matched model answer. In Table 4, we see that the majority of Davinci users have comparable confidence levels and agree with their model on 53% of the cases. Therefore only for 10% of the questions, users prefer other models’ answer with the same label. However, overall they have the lowest agreement with their model. For RoBERTa, aligned users choose higher confident model answers in 40% of the cases. RoBERTa has the lowest average confidence and its users tend to select more confident models. As for LSTM, the agreement between users and the model is nearly the same as with RoBERTa. Nevertheless, in most cases, people choose the LSTM model itself, but in 14% of the cases, they choose models with lower confidence.

(3) What is the relation between collaboration and cognitive trust? We calculate Pearson’s r between collaboration and cognitive trust ratings of users. We first omit two outliers from the data. Then, over 98 user data ratings, with $r=0.3$, $p=0.002$, we see a weak positive correlation between them. This result once again suggests that cognitive trust may not be a necessity for high collaboration rates.

7 Related Work

Factors in human-AI collaboration: The factors to consider in HAI can change based on the type of AI-enabled systems, such

as whether it is a physical machine, such as a robot, or a virtual agent like today’s popular NLP models [17, 23]. Araujo et al. study the human factor by analyzing different user groups in automated decision-making with AI [2]. They consider privacy concerns, age, decision-making type, AI role, and other features and investigate AI’s perceived justice, usefulness, and risk. [6, 17, 23] study the relationship between trust and human-AI collaboration. Glikson and Woolley reviewed the trust factors and identified two dimensions of trust: i) the *cognitive trust* based on the “good rationality skills” of the object of trust, and ii) the *emotional trust* based on the positive effect for the object of trust [18, 24]. Another line of work studies AI-advised decision-making scenarios. For instance, Bansal et al. find that accuracy is not the primary metric for achieving the best human human-machine team performance [4]. Bansal et al. further optimize AI models with human preferences and suggests that models should minimize the number of solutions that provide low-confidence accurate answers in favor of high-confidence accurate answers for better collaboration [3]. Vodrahalli et al. report similar findings and show that high confidence in AI predictions increases human intention to incorporate them in their decision-making [39]. Moreover, Bansal et al. demonstrate that human awareness of AI errors enhances overall human-AI team accuracy. Thus, establishing clear and predictable AI error boundaries is crucial for improving collaboration with AI [4].

Soft-labels: Pavlick and Kwiatkowski demonstrate that the variation of human annotations in the SNLI dataset might contain valuable signals and should not be treated as noise [32]. It is a pattern observed when the number of annotators increases and can be identified within Gaussian mixture models. Plank, Uma et al. report the annotation variation and argue that this variation should not be discarded [34, 38]. Furthermore, this variation has been shown beneficial for optimizing models by using *soft labels*—probability distributions over human annotations—to achieve better performance with multi-task learning setups instead of single *hard labels* [14, 16, 27].

Sample size in user studies: In Human-Computer Interaction literature on human factors, a sample size of 100 is commonly applicable by local standards [11], where many studies involve

even fewer than 100 participants. For instance, Pinski et al. surveyed 111 participants to explore the impact of AI knowledge on enhancing AI delegation [33]. Similarly, Xu and Zhang examined the efficacy of AI augmentation for real-time mathematical tasks, assessing user anxiety through a Likert scale with a cohort of 80 participants [43]. Tretter et al. involved 81 participants to study AI's acceptance in human decision-makings [37]. Boonprakong et al. investigates the physiological effects of cognitive biases with 33 participants [8]. In the context of designing an editing tool for conceptual diagrams, Pan et al. engaged 44 participants [31], while Xu et al. study involved 78 individuals to compare AI recommendations in human-AI teamwork [42]. Furthermore, the ChaosNLI dataset [29] collected opinions from 100 participants to handle the diversity in human reasoning. Finally, Conroy et al. show that a sample size of 100 can provide a $\pm 10\%$ margin of uncertainty when studying group characteristics or behavior [13].

8 Conclusion and Future Work

This work presents a novel experimental setup to analyze decision-making similarities (DMS) between AI agents and humans, focusing on their influence on collaboration and cognitive trust. By leveraging divergence metrics such as α -KL, β -KL, and JSD, we find that collaboration is strongly linked to decision-making similarity, while cognitive trust depends on additional factors like avoiding overconfidence in uncertain cases. Future work will explore fine-tuning models to optimize for collaboration or trust and extend the proposed framework to diverse tasks, datasets, and user groups. The identified metrics can also serve as evaluation tools before deploying AI systems.

Limitations

This study has several limitations. The user study design results in uneven feedback across models, limiting conclusions for underrepresented cases. Its focus on a single task, SNLI-based natural language inference, restricts generalizability to other contexts. Conducted in a controlled environment with self-reported trust measures from a single session, the study does not capture trust evolution in real-world settings. Additionally, it does not account for participants' prior experience with AI, which may influence results. Addressing these issues can enhance robustness and applicability.

Ethics Statement

This study analyzes human perceptions of NLP models using data from 100 participants recruited via Prolific. Demographic data such as age, education, gender, and native language were collected anonymously, without any personally identifiable information. Results are shared publicly, and no specific claims are made about participants or groups.

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References

- [1] Rania Abdelghani, Yen-Hsiang Wang, Xingdi Yuan, Tong Wang, Pauline Lucas, Hélène Sauzéon, and Pierre-Yves Oudeyer. 2024. GPT-3-Driven Pedagogical Agents to Train Children's Curious Question-Asking Skills. *International Journal of Artificial Intelligence in Education* 34, 2 (01 Jun 2024), 483–518. doi:10.1007/s40593-023-00340-7
- [2] Theo Araujo, Natali Helberger, Sanne Kruikemeier, and Claes H De Vreese. 2020. In AI we trust? Perceptions about automated decision-making by artificial intelligence. *AI & society* 35 (2020), 611–623.
- [3] Gagan Bansal, Besmira Nushi, Ece Kamar, Eric Horvitz, and Daniel S Weld. 2021. Is the most accurate AI the best teammate? Optimizing AI for teamwork. *Proc. Conf. AAAI Artif. Intell.* 35, 13 (May 2021), 11405–11414.
- [4] Gagan Bansal, Besmira Nushi, Ece Kamar, Walter S. Lasecki, Daniel S. Weld, and Eric Horvitz. 2019. Beyond Accuracy: The Role of Mental Models in Human-AI Team Performance. *Proceedings of the AAAI Conference on Human Computation and Crowdsourcing* 7, 1 (Oct. 2019), 2–11. doi:10.1609/hcomp.v7i1.5285
- [5] Gagan Bansal, Tongshuang Wu, Joyce Zhou, Raymond Fok, Besmira Nushi, Ece Kamar, Marco Tulio Ribeiro, and Daniel S. Weld. 2021. Does the Whole Exceed its Parts? The Effect of AI Explanations on Complementary Team Performance. arXiv:2006.14779 [cs.AI]
- [6] Ying Bao, Xusen Cheng, Triparna De Vreede, and Gert-Jan De Vreede. 2021. Investigating the relationship between AI and trust in human-AI collaboration. In *Proceedings of the 54th Hawaii International Conference on System Sciences*. Scholar Space, Grand Wailea, Maui, Hawaii, USA, 607–616.
- [7] Shraddha Barke, Michael B James, and Nadia Polikarpova. 2023. Grounded copilot: How programmers interact with code-generating models. *Proceedings of the ACM on Programming Languages* 7, OOPSLA1 (2023), 85–111.
- [8] Nattapat Boonprakong, Xiuge Chen, Catherine Davey, Benjamin Tag, and Tilman Dingler. 2023. Bias-Aware Systems: Exploring Indicators for the Occurrences of Cognitive Biases When Facing Different Opinions. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 27, 19 pages. doi:10.1145/3544548.3580917
- [9] Samuel R. Bowman, Gabor Angeli, Christopher Potts, and Christopher D. Manning. 2015. A large annotated corpus for learning natural language inference. In *Proceedings of the 2015 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Lisbon, Portugal, 632–642. doi:10.18653/v1/D15-1075
- [10] Tom B. Brown, Benjamin Mann, et al. 2020. Language Models are Few-Shot Learners. arXiv:2005.14165 [cs.CL]
- [11] Kelly Caine. 2016. Local Standards for Sample Size at CHI. In *Proceedings of the 2016 CHI Conference on Human Factors in Computing Systems* (San Jose, California, USA) (CHI '16). Association for Computing Machinery, New York, NY, USA, 981–992. doi:10.1145/2858036.2858498
- [12] Qian Chen, Xiaodan Zhu, Zhen-Hua Ling, Si Wei, Hui Jiang, and Diana Inkpen. 2017. Enhanced LSTM for Natural Language Inference. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, Regina Barzilay and Min-Yen Kan (Eds.). Association for Computational Linguistics, Vancouver, Canada, 1657–1668. doi:10.18653/v1/P17-1152
- [13] Ronán M Conroy et al. 2016. The RCSI Sample size handbook. *A rough guide* 1, 1 (2016), 59–61.
- [14] Aida Mostafazadeh Davani, Mark Díaz, and Vinodkumar Prabhakaran. 2022. Dealing with Disagreements: Looking Beyond the Majority Vote in Subjective Annotations. *Transactions of the Association for Computational Linguistics* 10 (2022), 92–110. doi:10.1162/tacl_a_00449
- [15] Ramon Dijkstra, Zülküf Genç, Subhadeep Kayal, Jaap Kamps, et al. 2022. Reading Comprehension Quiz Generation using Generative Pre-trained Transformers.
- [16] Tommaso Fornaciari, Alexandra Uma, Silviu Paun, Barbara Plank, Dirk Hovy, and Massimo Poesio. 2021. Beyond Black & White: Leveraging Annotator Disagreement via Soft-Label Multi-Task Learning. In *Proceedings of the 2021 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*. Association for Computational Linguistics, Online, 2591–2597. doi:10.18653/v1/2021.naacl-main.204
- [17] Omri Gillath, Ting Ai, Michael S Branicky, Shawn Keshmiri, Robert B Davidson, and Ryan Spaulding. 2021. Attachment and trust in artificial intelligence. *Computers in Human Behavior* 115 (2021), 106607.
- [18] Ella Glikson and Anita Williams Woolley. 2020. Human trust in artificial intelligence: Review of empirical research. *Academy of Management Annals* 14, 2 (2020), 627–660.
- [19] Alex Howard, William Hope, and Alessandro Gerada. 2023. ChatGPT and antimicrobial advice: the end of the consulting infection doctor? *The Lancet Infectious Diseases* 23, 4 (2023), 405–406.

- [20] Katharina Jeblick, Balthasar Schachtner, Jakob Dextl, Andreas Mittermeier, Anna Theresa Stüber, Johanna Topalis, Tobias Weber, Philipp Wesp, Bastian Oliver Sabel, Jens Ricke, and Michael Ingrisch. 2024. ChatGPT makes medicine easy to swallow: an exploratory case study on simplified radiology reports. *European Radiology* 34, 5 (01 May 2024), 2817–2825. doi:10.1007/s00330-023-10213-1
- [21] Rafal Kocielnik, Saleema Amershi, and Paul N. Bennett. 2019. Will You Accept an Imperfect AI? Exploring Designs for Adjusting End-user Expectations of AI Systems. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–14. doi:10.1145/3290605.3300641
- [22] Solomon Kullback and Richard A. Leibler. 1951. On information and sufficiency. *The annals of mathematical statistics* 22, 1 (1951), 79–86.
- [23] Min Kyung Lee. 2018. Understanding perception of algorithmic decisions: Fairness, trust, and emotion in response to algorithmic management. *Big Data & Society* 5, 1 (2018), 2053951718756684.
- [24] J. David Lewis and Andrew Weigert. 1985. Trust as a social reality. *Social forces* 63, 4 (1985), 967–985.
- [25] Jianhua Lin. 1991. Divergence measures based on the Shannon entropy. *IEEE Trans. Inf. Theory* 37 (1991), 145–151.
- [26] Jialin Liu, Changyu Wang, and Siru Liu. 2023. Utility of ChatGPT in clinical practice. *Journal of Medical Internet Research* 25 (2023), e48568.
- [27] Johannes Mario Meissner, Napat Thumwanit, Saku Sugawara, and Akiko Aizawa. 2021. Embracing Ambiguity: Shifting the Training Target of NLI Models. In *Proceedings of the 59th Annual Meeting of the Association for Computational Linguistics and the 11th International Joint Conference on Natural Language Processing (Volume 2: Short Papers)*, Chengqing Zong, Fei Xia, Wenjie Li, and Roberto Navigli (Eds.). Association for Computational Linguistics, Online, 862–869. doi:10.18653/v1/2021.acl-short.109
- [28] Steven Moore, Huy A. Nguyen, Norman Bier, Tanvi Domadia, and John Stamper. 2022. Assessing the Quality of Student-Generated Short Answer Questions Using GPT-3. In *Educating for a New Future: Making Sense of Technology-Enhanced Learning Adoption: 17th European Conference on Technology Enhanced Learning, EC-TEL 2022, Toulouse, France, September 12–16, 2022, Proceedings* (Toulouse, France). Springer-Verlag, Berlin, Heidelberg, 243–257. doi:10.1007/978-3-031-16290-9_18
- [29] Yixin Nie, Xiang Zhou, and Mohit Bansal. 2020. What Can We Learn from Collective Human Opinions on Natural Language Inference Data?. In *Proceedings of the 2020 Conference on Empirical Methods in Natural Language Processing (EMNLP)*. Association for Computational Linguistics, Online, 9131–9143. doi:10.18653/v1/2020.emnlp-main.734
- [30] OpenAI. 2022. Introducing ChatGPT. <https://openai.com/blog/chatgpt>
- [31] Lihang Pan, Chun Yu, Zhe He, and Yuanchun Shi. 2023. A Human-Computer Collaborative Editing Tool for Conceptual Diagrams. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 360, 29 pages. doi:10.1145/3544548.3580676
- [32] Ellie Pavlick and Tom Kwiatkowski. 2019. Inherent Disagreements in Human Textual Inferences. *Transactions of the Association for Computational Linguistics* 7 (2019), 677–694. doi:10.1162/tacl_a_00293
- [33] Marc Pinski, Martin Adam, and Alexander Benlian. 2023. AI Knowledge: Improving AI Delegation through Human Enablement. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 25, 17 pages. doi:10.1145/3544548.3580794
- [34] Barbara Plank. 2022. The “Problem” of Human Label Variation: On Ground Truth in Data, Modeling and Evaluation. In *Proceedings of the 2022 Conference on Empirical Methods in Natural Language Processing*. Association for Computational Linguistics, Abu Dhabi, United Arab Emirates, 10671–10682. <https://aclanthology.org/2022.emnlp-main.731>
- [35] Donghee Shin. 2021. The effects of explainability and causability on perception, trust, and acceptance: Implications for explainable AI. *International Journal of Human-Computer Studies* 146 (2021), 102551. doi:10.1016/j.ijhcs.2020.102551
- [36] Marita Skjuve, Petter Bae Brandtzaeg, and Asbjørn Følstad. 2024. Why do people use ChatGPT? Exploring user motivations for generative conversational AI. *First Monday* 29, 1 (Jan. 2024), 23. doi:10.5210/fm.v29i1.13541
- [37] Stefan Tretter, Axel Platz, and Sarah Diefenbach. 2023. Matching Mind and Method: Augmented Decision-Making with Digital Companions Based on Regulatory Mode Theory. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 28, 10 pages. doi:10.1145/3544548.3581529
- [38] Alexandra N Uma, Tommaso Fornaciari, Dirk Hovy, Silviu Paun, Barbara Plank, and Massimo Poesio. 2021. Learning from disagreement: A survey. *Journal of Artificial Intelligence Research* 72 (2021), 1385–1470.
- [39] Kailas Vodrahalli, Tobias Gerstenberg, and James Y. Zou. 2022. Uncalibrated models can improve human-ai collaboration. *Advances in Neural Information Processing Systems* 35 (2022), 4004–4016.
- [40] Alex Wang, Yada Pruksachatkun, Nikita Nangia, Amanpreet Singh, Julian Michael, Felix Hill, Omer Levy, and Samuel R. Bowman. 2019. *SuperGLUE: a stickier benchmark for general-purpose language understanding systems*. Curran Associates Inc., Red Hook, NY, USA, Chapter 1, 3266–3280.
- [41] Albert Webson and Ellie Pavlick. 2022. Do Prompt-Based Models Really Understand the Meaning of their Prompts? arXiv:2109.01247 [cs.CL]
- [42] Chengyuan Xu, Kuo-Chin Lien, and Tobias Höllerer. 2023. Comparing Zealous and Restrained AI Recommendations in a Real-World Human-AI Collaboration Task. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 350, 15 pages. doi:10.1145/3544548.3581282
- [43] Songlin Xu and Xinyu Zhang. 2023. Augmenting Human Cognition with an AI-Mediated Intelligent Visual Feedback. In *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems* (Hamburg, Germany) (CHI '23). Association for Computing Machinery, New York, NY, USA, Article 26, 16 pages. doi:10.1145/3544548.3580905
- [44] Ming Yin, Jennifer Wortman Vaughan, and Hanna Wallach. 2019. Understanding the Effect of Accuracy on Trust in Machine Learning Models. In *Proceedings of the 2019 CHI Conference on Human Factors in Computing Systems* (Glasgow, Scotland UK) (CHI '19). Association for Computing Machinery, New York, NY, USA, 1–12. doi:10.1145/3290605.3300509
- [45] Yunfeng Zhang, Q. Vera Liao, and Rachel K. E. Bellamy. 2020. Effect of confidence and explanation on accuracy and trust calibration in AI-assisted decision making. In *Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency* (Barcelona, Spain) (FAT* '20). Association for Computing Machinery, New York, NY, USA, 295–305. doi:10.1145/3351095.3372852
- [46] Liu Zhuang, Lin Wayne, Shi Ya, and Zhao Jun. 2021. A Robustly Optimized BERT Pre-training Approach with Post-training. In *Proceedings of the 20th Chinese National Conference on Computational Linguistics*, Sheng Li, Maosong Sun, Yang Liu, Hua Wu, Kang Liu, Wanxiang Che, Shizhu He, and Gaoqi Rao (Eds.). Chinese Information Processing Society of China, Huhhot, China, 1218–1227. <https://aclanthology.org/2021.ccl-1.108/>

A Glossary of Terms

Term	Definition/Explanation
Soft Labels	Probability distributions over human annotations that reflect the degree of confidence or consensus among annotators, as opposed to a single categorical label.
3GS/4GS/5GS	Notations indicating the level of agreement among annotators. For example, 3GS means that up to three annotators agreed on the label, whereas 5GS indicates unanimous agreement.
KL Divergence	Kullback-Leibler divergence; a measure of how one probability distribution diverges from another expected distribution.
JSD (Jensen-Shannon Divergence)	A symmetric and finite divergence measure derived from KL divergence, which quantifies the similarity between two probability distributions.
Decision-Making Similarity (DMS)	A metric that quantifies the similarity between human and AI decision processes by comparing their soft label distributions using divergence metrics.

Table 5: Glossary of key terms used in this paper.

B Details of Models

Random baseline: It assigns probabilities, a.k.a., confidence scores, randomly to each label (i.e., neutral, contradiction, entailment) drawn from a uniform distribution. A small Gaussian noise with a mean of 0 and variance of 0.05 is added to the soft labels⁷ and then normalized for the user study to avoid easy recognition.

TF-IDF: We extract TF-IDF weighted lexical features from both premise and hypothesis sentences and train a statistical linear classifier using the LBFGS optimizer.

RoBERTa [46]: We fine-tune the roberta-large for pair-wise classification on SNLI training dataset by excluding the instances that are demonstrated to human annotators. We use the default parameters provided with AllenNLP.

Enhanced LSTM [12]: The model uses a chain of LSTMs. The first layer, a bi-LSTM, encodes the premise and hypothesis pairs. The model then applies a form of soft alignment between premise and hypothesis token pairs to enhance the representations. Finally, the representations are fed into another bi-LSTM, and then passed through a pooling operation, followed by a softmax. We train the model from scratch with the default parameters.

da-vinci-003 [10]: We do zero-shot prompting with text-davinci-003 model through its publicly available API and set max_token to 1. We use the prompt template similar to [41] given in Table 6. We

⁷We use soft label, confidence scores, and probabilities interchangeably throughout the paper.

then map the model’s responses of “yes”, “maybe” and “no” to the entailment, neutral, and contradiction labels, respectively.

```
{premise_sentence}
Can we infer that {hypothesis_sentence}?
Answer as yes, no, or maybe. Answer:
```

Table 6: Prompt template for text-davinci-003.

C Phases of the User Study

i) Training Phase: Users are provided with guidelines, which contain the general flow of the study with detailed instructions and a detailed explanation of the annotation task with examples. Full annotation guidelines can be found in Table 7.

ii) Quiz Phase: Participants answer ten questions and then see the correct answers with explanations for training purposes in the quiz phase.

iii) Annotation Phase: The participants are shown the selected premise-hypothesis pairs, along with the label distributions from the selected models (see §2.2) as shown in Fig 2. The users are instructed to choose the *closest agent* by means of the label distributions. The order of the models is shuffled each time to prevent users from finding shortcuts. Only one option can be chosen. Participants can skip the question and go back to the question. They are also asked to rate the difficulty of the question from 1 to 10.

After the annotation phase, we calculate the most similar model to the user with the JSD metric. This is because JSD contains both the α and the β -KL calculations, penalizing models that are overly confident on a single label w.r.t the users (high α -KL), and vice versa (high β -KL). We calculate the distance between the user’s answer and all the other label distributions. Then, we sum the distances for each model and identify the most similar one to the user, which we refer to as the *aligned model*.

Premise: A little boy is opening gifts surrounded by a group of children and adults.

Hypothesis: The boy is being punished

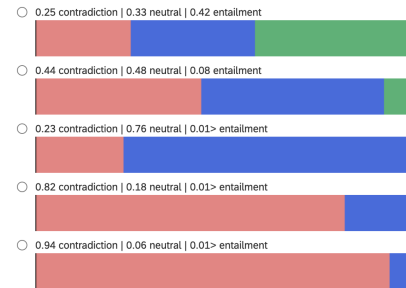


Figure 2: The annotation task is framed as a multiple-choice question answering problem, where the available options correspond to label predictions generated by the selected models.

User Study Guidelines

Page 1.

In this survey, you will be shown a pair of sentences in which you should determine the inference relation between them. The first sentence is a premise, and the second is a candidate hypothesis. There are three possible inference relations between sentences: entailment, neutral, or contradiction.

In the first part of the survey, you will be asked 10 test questions to confirm the task is understood. In the second part, you will be asked 50 questions from the task. In the final part, you will be asked task-related questions. The survey will take approximately 30 min.

Thank you for your participation.

Page 2.

Task Examples:

Entailment: Premise sentence definitely implies the hypothesis.
e.g.:

Premise: “Two dogs are running through a field.”

Hypothesis: “There are animals outdoors.”

Correct label: Entailment

Neutral: Premise sentence might imply the hypothesis.

e.g.:

Premise: “Two dogs are running through a field.”

Hypothesis: “Some puppies are running to catch a stick”

Correct label: Neutral

Contradiction: Premise sentence definitely does not imply the hypothesis.

e.g.:

Premise: “Two dogs are running through a field.”

Hypothesis: “The pets are sitting on a couch.”

Correct label: Contradiction. This is different from the *maybe correct (neutral)* category because it’s impossible for the dogs to be both running and sitting.

Page 3.

In the next part of the survey, you will answer 50 questions with multiple choices that include confidence-relation pairs. You can select the choice which reflects your confidence in labels.

Premise: premise_sentence

Hypothesis: hypothesis_sentence

Choice format:

x% Contradiction | y% Neutral | z% Entailment

e.g.:

0.55 Contradiction | 0.35 Neutral | 0.10 Entailment

Table 7: User study guideline

iv) Dynamic Survey Phase: In the dynamic survey phase, the users are asked to review their own selections side-by-side with their aligned model as shown in Figure 3. They can investigate four categories: (A) agreements with the aligned model: the common successes and failures, and (B) the disagreements with the aligned model: the cases where the model succeeded and the user failed, and vice versa.

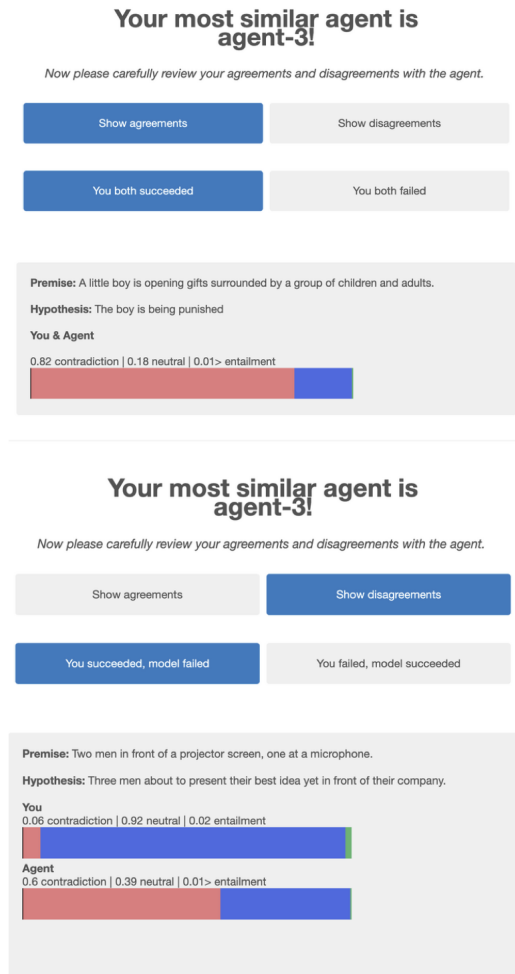


Figure 3: During the dynamic survey phase, users are prompted to assess their own selections alongside their aligned model. This involves exploring four categories: agreements with the aligned model, which encompass common successes and failures (top), and disagreements with the aligned model, which include instances where the model succeeded while the user failed, and vice versa (bottom).

After they revise the similarities and dissimilarities, they are asked several questions regarding the model behavior and the following outcomes: (1) Collaboration: “Would you collaborate with this agent to accomplish the task?”, (2) Cognitive Trust: “On a scale of 1 to 5, how much do you trust the agent to make rational decisions?” For collaboration, the choices are provided in verbal likert scale format, including “definitely not”, “probably not”, “neutral”, “probably yes”, and “definitely yes”. Finally, to validate our assumptions of the task, we ask several task-related questions, such as the difficulty and subjectivity of the task, and skills needed to solve the task, such as emotional awareness, logical thinking, and being a

native English speaker. To date, we have not exploited this information, and we will do so in future work, especially when there are more than SNLI tasks to perform.

D Distributions of Collaboration and Cognitive Trust

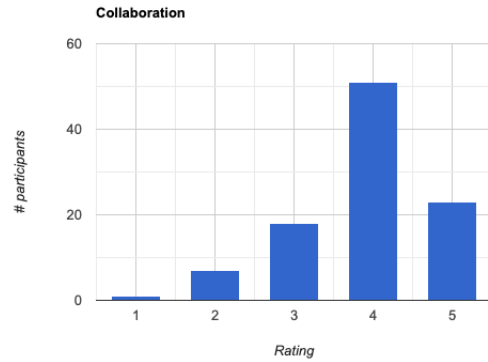


Figure 4: Collaboration ratings among users. Over half of the participants give 4/5 ratings for collaboration with their aligned model.

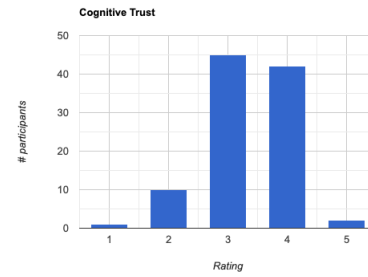


Figure 5: Cognitive Trust ratings among users. Participants are distributed across 3 and 4/5 ratings for their aligned model.